

How to Choose the Right ML Model

12 Cases | 24 Real Examples | Visual Flow for Every Decision

Each example walks through the 5 questions step by step

Step 0: From Problem to Data — "I Have No Data Yet"

You don't need data to CHOOSE the model. You need the PROBLEM.



Answer Type Tells You What Data to Collect:

Answer Type	Data Needed	Labels?
WORD (spam/not)	Labeled examples	YES
NUMBER (Rs 72L)	Past records with numbers	YES
GROUPS (types)	Just raw data	NO
WEIRD (anomaly)	Normal examples only	NO
STRATEGY (RL)	Simulator / environment	NO

How Much Data Do You Need?

Model	Minimum	Ideal
Logistic Reg / DT	500	5K+
Random Forest	1K	10K+
CNN from scratch	10K imgs	100K+
CNN + Transfer Learn	200 imgs	1K+
K-Means / DBSCAN	500	5K+
Autoencoder	1K normal	10K+
RL	1M simulated	More=better

ExamGuard Phones

Phone/No Phone = WORD
Need labeled IMAGES
Take 500 photos + augment
YOLO + Transfer Learning

Predict Sales

Rs amount = NUMBER
Need past sales records
Export from database (60 months)
Linear Regression first

Customer Groups

Groups = CLUSTERING
Need customer info
Export 50K from CRM
K-Means (K=3)

The 5 Questions You Ask Every Time

Q1: Answer Type?

WORD = Classification
NUMBER = Regression
GROUPS = Clustering
STRATEGY = RL

Q2: Have Labels?

YES = Supervised
NO = Unsupervised
REWARDS = RL

Q3: Data Type?

IMAGES = CNN/YOLO
TEXT = Naive Bayes
NUMBERS = Many options

Q4: How Much Data?

<5K = Simple models
5K-100K = Most models
100K+ = Deep Learning

Q5: Special Needs?

Explain? = Decision Tree
Real-time? = YOLO
Imbalanced? = Anomaly

Classification Master Flow — "Answer is a WORD"



Regression Master Flow — "Answer is a NUMBER"

1. ALWAYS Linear First

Try simplest
Check R-squared

$R^2 > 0.80$?

Good enough?
YES = DONE!

Data CURVED?

Plot it.
Bends? = YES

Many features
(10+)?

Complex
interactions?

MASSIVE
data (100K+)?

50+ features?

v

LINEAR REG
 $R^2=0.85$ DONE!

v

POLYNOMIAL
Fits curves!

v

RF REGRESSION
100 trees average

v

NEURAL NET
Most powerful

THE RULE: Linear >>> Polynomial >>> RF Regression >>> Neural Net. NEVER skip ahead. Always start simple.

Unsupervised + RL Flow — "No Labels / Strategy"

FIND GROUPS (Clustering)

Know how many groups?

YES >>> K-MEANS (you pick K)

NO >>> DBSCAN (finds K + outliers)

Not sure? Elbow Method for K

FIND WEIRD (Anomaly)

What type of data?

NUMBERS >>> Isolation Forest

IMAGES >>> Autoencoder

Only know normal? >>> Autoencoder

LEARN STRATEGY (RL)

3 conditions needed:

1. Sequential decisions
2. Actions affect future
3. Can define rewards

Strategy=RL | Balance alerts=RL

Case 1: Logistic Regression

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
Email Spam Filter 5K emails, labeled spam / not-spam	Answer? WORD Spam/Not	Labels? YES 5K labeled	Data? NUMBERS word count,links	Special? NO explain small data	LOGISTIC REGRESSION	CNN: not images RF: overkill for 5K DT: no explain need
Employee Quit 6K records, labeled quit / stayed	Answer? WORD Quit/Stay	Labels? YES 6K labeled	Data? NUMBERS salary,years,hrs	Special? Need probability score for HR	LOGISTIC REGRESSION	NN: 6K too few DT: HR uses internally Outputs probability %

START SIMPLE: Logistic Regression = fastest baseline. Only upgrade if accuracy < 80%. Perfect for 2 categories + small data.

Case 2: Decision Tree

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
Diabetes Diagnosis 15K patients age,BP,sugar labeled Y/N	Answer? WORD Yes / No	Labels? YES 15K labeled	Data? NUMBERS age,sugar,BP	Special? Doctor MUST EXPLAIN why	DECISION TREE	RF: 88% vs DT 82% but can't explain 100 trees to patient
Car Insurance 30K policies customer asks WHY premium high	Answer? WORD Low/Med/High	Labels? YES 30K labeled	Data? NUMBERS age,accidents,city	Special? MUST explain to angry customer	DECISION TREE	LogReg: can't show WHICH factor caused high premium

USE WHEN EXPLANATION MATTERS: Bank tells customer WHY loan rejected. Doctor explains WHY diagnosis. 6% less accuracy but legally defensible.

Case 3: Random Forest

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
Disease Diagnosis 100K patients need BEST accuracy	Answer? WORD disease type	Labels? YES 100K labeled	Data? NUMBERS 30 features	Special? BEST accuracy no explain needed	RANDOM FOREST	DT: 82% accuracy RF: 89% accuracy 100 trees > 1 tree
Product Defects 50K products predict defective/ok	Answer? WORD Defect/OK	Labels? YES 50K labeled	Data? NUMBERS 20 features	Special? Accuracy over everything else	RANDOM FOREST	LogReg: too simple for 20 features RF handles complexity

BEST ACCURACY ON TABLES: 100 trees vote = almost always wins. Use when you DON'T need to explain and ACCURACY is king.

Case 4: CNN / YOLO

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
ExamGuard Phones 500 labeled photos phone / no phone	Answer? WORD Phone/None	Labels? YES 500 images	Data? IMAGES! Camera frames	Special? Real-time 30fps only 500 images	YOLO + Transfer Learn	RF: can't see pixels CNN scratch: 500 few YOLO pre-trained OK
Traffic Violations Camera footage red-light violations	Answer? WORD Violation/OK	Labels? YES labeled clips	Data? IMAGES! Traffic camera	Special? Real-time 24/7 monitoring	YOLO 30+ fps	Only CNN handles images. YOLO fastest. No other option.

IMAGES = CNN. No other model can process pixels. <500 images? Transfer Learning. Real-time? YOLO. No exceptions.

Case 5: The Imbalance Trap

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
Credit Card Fraud 1M transactions 99.9% normal, 0.1% fraud	Answer? WORD Fraud/Normal	Labels? YES but 99.9% / 0.1%!	Data? NUMBERS amount,time,loc	TRAP! IMBALANCED! DON'T classify!	ISOLATION FOREST	LogReg: says 'normal' every time = 99.9% acc, 0% fraud caught!
Factory Defects 100K items 99.5% ok, 0.5% defective	Answer? WORD Defect/OK	Labels? YES but 99.5% / 0.5%!	Data? NUMBERS sensor readings	TRAP! IMBALANCED! DON'T classify!	ISOLATION FOREST	RF: memorizes 'ok' misses rare defects Need anomaly approach

THE TRAP: 99.9% accuracy means NOTHING if 0% fraud caught! When data is 95/5 or worse, STOP classifying. Use Anomaly Detection.

Case 6: Naive Bayes

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
Product Reviews 20K reviews labeled positive / negative	Answer? WORD Pos / Neg	Labels? YES 20K labeled	Data? TEXT! Review words	Special? Small text data word frequencies	NAIVE BAYES	CNN: not images LogReg: NB designed for text specifically
Support Tickets 10K tickets route to department	Answer? WORD Sales/Tech/Bill	Labels? YES 10K labeled	Data? TEXT! Ticket content	Special? Multiple categories text classification	NAIVE BAYES	BERT: 10K too small for transformer NB fast + effective

TEXT DATA = NAIVE BAYES first. Designed for word frequencies. Fast, simple, surprisingly effective. Only upgrade to BERT for large + context-heavy text.

Case 7: Linear Regression

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
House Prices 10K houses predict price in Rs	Answer? NUMBER Rs 72 lakhs	Labels? YES sold prices	Try Linear? $R^2 = 0.85$ Good enough!	Shape? Straight line Bigger=costlier	LINEAR REGRESSION	Poly: not curved RF: overkill, Linear already 85%
Salary Predict 8K employees predict monthly salary	Answer? NUMBER Rs/month	Labels? YES actual salaries	Try Linear? $R^2 = 0.82$ Good enough!	Shape? Experience up = salary up	LINEAR REGRESSION	Neural: 8K too small Linear works fine Always start simple

ALWAYS START LINEAR: If $R^2 > 0.80$, you're DONE. Don't overcomplicate. Upgrade ONLY when accuracy demands it.

Case 8: Polynomial Regression

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
Crop Yield 5K farms fertilizer vs yield	Answer? NUMBER kg / acre	Try Linear? R² = 0.52 BAD!	Plot data? CURVED! Up then down	Why curved? Too much fert = crop dies	POLYNOMIAL REGRESSION	Linear: R ² only 0.52 misses the curve Poly R ² = 0.91
Ad Spend ROI 3K campaigns spend vs revenue	Answer? NUMBER revenue Rs	Try Linear? R² = 0.48 BAD!	Plot data? CURVED! Diminishing returns	Why curved? More spend helps then wastes money	POLYNOMIAL REGRESSION	Linear: straight line can't show diminishing returns = curve

CURVED DATA = POLYNOMIAL: If Linear R² is bad, PLOT the data. See a curve? Polynomial fixes it. Too much of something = diminishing returns = CURVE.

Case 9: K-Means vs DBSCAN

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
Customer Groups 50K customers marketing wants 3 segments	Labels? NO labels no type assigned	Find what? GROUPS similar together	Know K? YES! Marketing wants exactly 3	Data? NUMBERS age,spend,visits	K-MEANS K = 3	DBSCAN: marketing NEEDS exactly 3 not 'whatever forms'
GPS Truck Routes 10K GPS points find route patterns	Labels? NO labels no route types	Find what? GROUPS similar routes	Know K? NO! Don't know how many types	Need? Also find outlier trucks	DBSCAN auto K + outliers	K-Means: must pick K but don't know it can't find outliers

Know how many groups? K-MEANS. Don't know? DBSCAN (finds groups + outliers automatically). Marketing says '3 segments' = K-Means. Exploring = DBSCAN.

Case 10: Autoencoder

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
ExamGuard Unusual Camera feeds no labels for 'unusual'	Labels? NO labels can't label every cheating type	Find what? WEIRD stuff not normal	Data? IMAGES/VIDEO camera frames	Know what? DON'T know what unusual looks like	AUTO- ENCODER	IF: for numbers not images CNN: needs labels! Learn normal, flag rest
Server Monitor Server logs detect anomalies	Labels? NO labels only know normal	Find what? WEIRD stuff server problems	Data? Complex patterns many metrics	Know what? Only know what 'normal' looks like	AUTO- ENCODER	IF: patterns too complex for tree Autoencoder learns normal deeply

UNKNOWN ANOMALIES: Train ONLY on normal data. Anything that doesn't match = flag it. Works for images (Autoencoder) and numbers (Isolation Forest).

Case 11: RL Strategy

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
Self-Driving Car Driving simulator brake/steer/accelerate	Answer? Not WORD or NUMBER	Type? STRATEGY over time	Simulate? YES! Millions of drives	Rewards? Safe: +1000 Crash: -10000	REINFORCEMENT LEARNING	Supervised: can't label billion driving scenarios. RL learns.
Warehouse Robot Pick/place route optimize path	Answer? Not WORD or NUMBER	Type? SEQUENCE of actions	Simulate? YES! Virtual warehouse	Rewards? Fast: +100 Drop item: -500	REINFORCEMENT LEARNING	Too many possible paths to label. RL discovers best route by trying.

RL = STRATEGY OVER TIME: Sequential decisions + actions affect future + can simulate millions of tries. Can't label everything? RL discovers the strategy.

Case 12: RL Alerts / Balance

PROBLEM	Q1	Q2	Q3	Q4	MODEL	WHY NOT
ExamGuard Alerts Balance false alarms vs missed cheating	Answer? Not just classify	Type? BALANCE competing goals	Context? Day/night student history	Rewards? Correct: +100 False: -50 Miss: -200	RL ALERT SYSTEM	Classification: same answer always. RL adapts to context and history.
ICU Monitor Balance alarm fatigue vs missing critical	Answer? Not just classify	Type? BALANCE alert levels	Context? Patient state time of day	Rewards? Save life: +1000 False alarm: -30 Miss critical: -5000	RL ALERT SYSTEM	Classification: alerts on everything=fatigue RL: learns WHEN to escalate vs gentle warn

RL for BALANCE: When you need to balance competing goals (alert vs false alarm), RL learns WHEN to act based on context. Classification gives same answer always.